# Aryan Mohan

**AI Engineer | Generative AI, LLM Agents & RAG on Google Cloud**
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## SUMMARY

AI Engineer building production generative AI applications, LLM agents and RAG architectures on Google Cloud (Gemini on Vertex AI, Vertex AI Vector Search, LangChain, serverless Cloud Run). Led a zero-downtime AWS-to-Google Cloud migration and moved 8 AI agent services from OpenAI to Gemini with security, governance and cost controls built in. Presents architecture, demos and trade-offs to investors and prospective clients.

## EDUCATION

**University of Petroleum and Energy Studies (UPES)** | *B.Tech, CSE (AI & ML)* | Dehradun **Aug 2021 – May 2025**

## EXPERIENCE

**EdviseMe** | *Full-Stack AI Engineer* | Remote, India **May 2026 – Present**

- **Gemini agent migration:** Led the migration of 8 AI agent services (tutoring chat, quizzes, memory, support, mock-test grading, analytics) from OpenAI to stateless Gemini on Vertex AI via a provider-agnostic LLM layer and session-safe phased rollout.
- **Enterprise RAG on Vertex AI:** Consolidated 81 OpenAI vector stores into one Vertex AI Vector Search index, avoiding 81 endpoints (∼$70–110/month each); embedded 75,621 chunks (∼60M chars) with gemini-embedding-2 in ∼2 hours for ∼$1.40.
- **Grounding LLMs in internal data:** Built a retrieval service replacing OpenAI file_search that joins vector hits to MongoDB source data, with a 20-message memory window; verified 100% ingestion and in-domain relevance of 0.73–0.79 vs. ∼0.51 for off-domain queries.
- **Troubleshooting agent behavior:** Fixed stateless "amnesia" in the quiz agent with per-turn state injection and auto-retry; moved a 12KB structured-output report schema from Gemini 2.5 Pro to 2.5 Flash, reaching 100% schema validation with lower latency and token cost.
- **Serverless cloud migration:** Ran the zero-downtime AWS-to-Google Cloud production cutover (scoped at 34 engineer-days, core work in <15): Cloud Run, Direct VPC Egress + Cloud NAT, Global Load Balancer, Cloud CDN, Cloud Build CI/CD, blue-green rollback.
- **Security, governance & compliance:** Set up least-privilege IAM, Secret Manager and credential rotation; RBAC-protected LLM token/cost telemetry across 10 AI call sites; child-safety moderation on Cloud Natural Language API (0 false negatives in back-tests).
- **Multimodal document AI:** Built a 7-stage Gemini + Document AI pipeline (OCR, LaTeX math, diagram cropping) for exam papers; ran a GPT-4o Vision audit of 6,368 records that caught a ∼55% data-integrity failure rate before it reached live AI grading.
- **LLM-as-a-Judge evaluation:** Turned an expensive, hallucination-prone "Solver" into a low-cost "Evaluator" (retrieval + chain-of-thought) that eliminated math and score errors, hardened with idempotent retries and an auto-heal sweep.
- **Stakeholder advisory & leadership:** Co-led an investor pitch and product demos; built a multi-tenant, white-label B2B prototype with role-based access for institutional pitches; ran a 2-hour architecture KT for incoming leadership; lead junior engineers.

**ARMAX Solutions** | *Founder & AI Architect (B2B GenAI Consultancy)* | Remote **May 2025 – Apr 2026**

- Founded a B2B generative-AI consultancy, working directly with clients to deliver LLM prototypes and MERN-stack architectures.

**HERE Technologies (Volkswagen Group)** | *Web Scraping Intern* | Remote **Jun 2024 – Jul 2024**

- Automated large-scale geospatial data extraction in Python (Scrapy, Requests) to improve location-service accuracy.

**Plasmid Innovation Ltd.** | *Data Science Intern* | Remote **May 2024 – Jun 2024**

- Built data-cleaning pipelines and predictive ML models (Pandas, Scikit-learn, NumPy) for global purchasing trends.

## PROJECTS

- **Intelligent Incident Parser:** Strict JSON-schema LLM parsing of unstructured system alerts at 1–2 s inference latency.
- **Weather Image Classification:** Fine-tuned PyTorch ResNet-101 to 92% accuracy across 11 classes (XGBoost + Genetic Algorithm).
- **Text-to-Image Generator:** Generative image pipeline in a 2026 GitHub portfolio spanning LangGraph agent workflows and AWS CI/CD.

## TECHNICAL SKILLS

**Generative AI & Agents:** Gemini 2.5 Pro / Flash on Vertex AI, OpenAI (GPT-4o, GPT-4o Vision, Responses API), LangChain (ChatVertexAI, structured output), LangGraph, multi-agent workflows, RAG, prompt engineering, LLM-as-a-Judge, chain-of-thought, guardrails, PyTorch

**Google Cloud:** Vertex AI, Vertex AI Vector Search, gemini-embedding-2, Document AI, Cloud Natural Language API, Cloud Run, Cloud Build, Artifact Registry, Cloud Storage, Secret Manager, IAM / service accounts, Direct VPC Egress, Cloud NAT, Global External Load Balancer, Cloud CDN, Cloud Scheduler, Firebase / Identity Platform

**AWS & DevOps:** ECS, S3, CloudFront, ALB, Lambda, API Gateway, DynamoDB, CodeBuild, CodePipeline, Secrets Manager; Docker, CI/CD, Git

**Data & Engineering:** Python, Node.js / Express, React 19, Next.js, REST, SSE, MongoDB Atlas (aggregations, BulkWrite, zero-downtime migrations), SQL, ETL, Pandas, Scikit-learn, OAuth / JWT / OTP, RBAC, HMAC webhooks, multi-tenant B2B SaaS

## CERTIFICATIONS

**Certification:** Microsoft Certified: Azure AI Engineer Associate (AI-102), 2024